DAYANANDA SAGAR UNIVERSITY

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SCHOOL OF ENGINEERING

Bachelor of Technology
in
Artificial Intelligence and Robotics

Major Project Phase-II Report

CATARACT PREDICTION USING VGG19, RESNET50, MOBILENETV2

Batch: 1

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CERTIFICATE

This is to certify that the Phase-II project work titled **"CATARACT PREDICTION USING VGG19, RESNET50, MOBILENETV2"** is carried out by **HARSHITHA P (ENG23RA0008), CHINTHA DEEPAK (ENG23RA0029), DHANUSH S (ENG23RA0030), ADHIBAN B (ENG23RA0025)** bonafide students of Bachelor of Technology in Computer Science and Engineering at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2024-2025**.

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Date: 29 / 05 /2025	Date: 29 / 05 /2025	Date: 29 / 05 /2025

DECLARATION

We, **HARSHITHA P (ENG23RA0008), CHINTHA DEEPAK (ENG23RA0029), DHANUSH S (ENG23RA0030), ADHIBAN B (ENG23RA0025)** are students of eighth semester B. Tech in **Computer Science and Engineering**, at School of Engineering, **Dayananda Sagar University**, hereby declare that the Major Project Stage-II titled **"CATARACT PREDICTION USING VGG19, RESNET50, MOBILENETV2"** has been carried out by us and submitted in partial fulfilment for the award of degree in **Bachelor of Technology in Artificial Intelligence and Robotics** during the academic year **2024-2025**.

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CHAPTER 1

INTRODUCTION

CHAPTER 1 INTRODUCTION

Cataract is one of the leading causes of vision impairment and blindness worldwide. It occurs due to clouding of the eye's natural lens, which leads to blurred vision and can progress to complete loss of sight if left untreated. Early detection and timely intervention are critical in preventing vision loss caused by cataracts. Traditional diagnosis methods rely heavily on clinical expertise and equipment, which might not be accessible in rural or underdeveloped regions. This project aims to develop an automated solution for cataract detection using deep learning techniques, specifically leveraging the VGG19 model for accurate image classification.

This project report outlines the problem, objective, methodology, and contributions of the proposed system. It is organized into several chapters, starting with the Introduction and ending with Summary and Conclusions. Each chapter is structured with specific sections and subsections to clearly convey the content and findings of the project.

1.1 INTRODUCTION:

This project focuses on the application of deep learning, particularly convolutional neural networks (CNNs), in the field of ophthalmology for cataract prediction. VGG19, a pre-trained deep learning model known for its performance in image classification tasks, is employed to analyse ocular images and classify them as cataract-affected or normal.

The goal of this project is to design, train, and validate a machine learning model that can assist in the early detection of cataracts through automated image analysis. The proposed system aims to enhance diagnostic accuracy, reduce human error, and provide a scalable solution that can be integrated into telemedicine platforms or mobile health applications.

1.1.1 SECOND LEVEL OF HEADING:

The title of Chapter 1 is "Introduction". It outlines the background of cataract-related issues, the motivation behind using artificial intelligence for medical diagnosis, and the objectives of the project. The scope of this work includes the development of a prediction model using VGG19, evaluation using performance metrics, and deployment in a user-friendly interface for practical use.

The chapter also briefly discusses the significance of deep learning in healthcare and how transfer learning helps in solving complex problems with limited datasets, which is often a challenge in the medical domain.

1.1.1.1 Third Level of Heading:

The next chapter, Literature Review, presents an overview of previous research work related to cataract detection and deep learning models. It critically analyzes the strengths and limitations of existing methods and establishes the need for the proposed approach. The emphasis is placed on identifying gaps in current systems and justifying the selection of VGG19 as the backbone of the model.

1.2. FIGURES AND TABLES

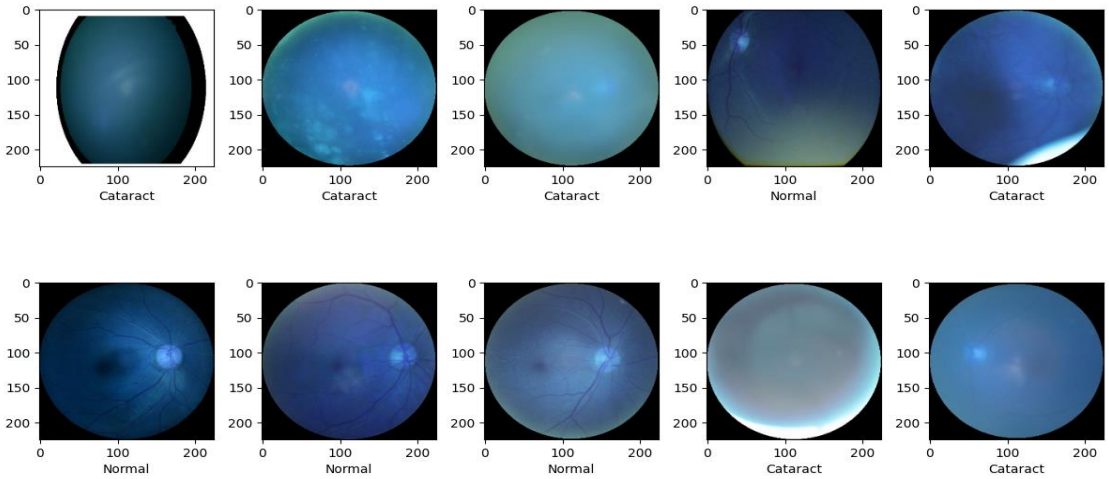


Figure 1 : Sample input

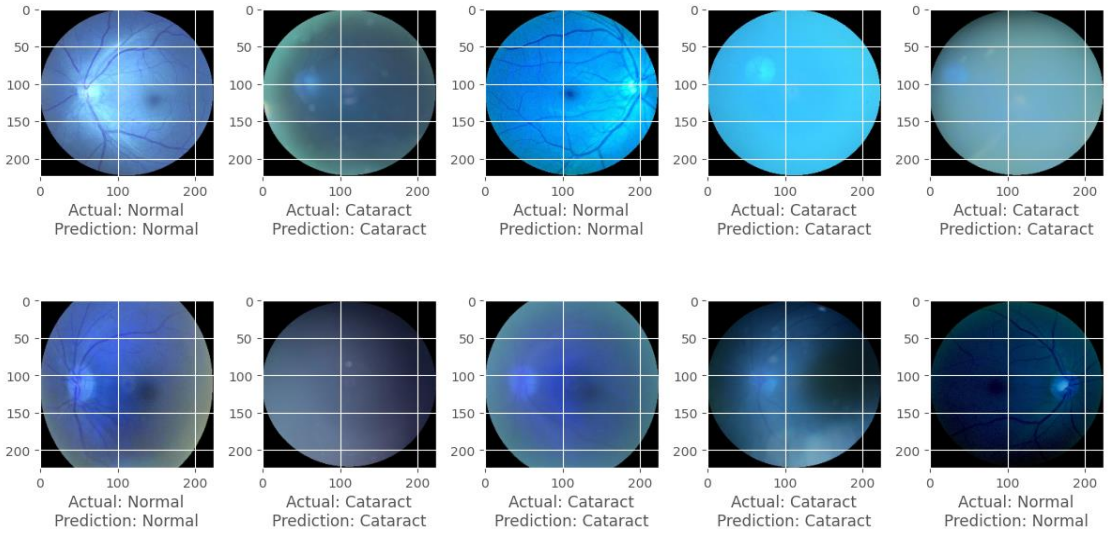


Figure 1.2 : Sample output

Table 1: Accuracy comparison

Model	Accuracy (%)	Comments
VGG19 (Transfer Learning)	97.16	Fine-tuned with additional dense layers, good generalization
ResNet50	96.75	Deeper architecture, risk of overfitting without regularization
MobileNetV2	94.10	Lightweight model for mobile deployment

1.3. SCOPE

The cataract prediction project holds significant potential to transform public health, especially in the field of ophthalmology, by addressing one of the leading causes of blindness worldwide. Cataracts are often left untreated in their early stages due to lack of awareness, limited access to healthcare, or the high cost of traditional diagnosis. This project aims to bridge these gaps through the use of technology, offering a scalable and accessible solution. One of the most profound social benefits is early detection, which allows for timely treatment and greatly reduces the risk of vision loss. Early diagnosis can significantly improve the quality of life, particularly for elderly individuals who are most susceptible to cataracts. By preserving their eyesight, individuals maintain their independence, mobility, and ability to contribute socially and economically within their families and communities. The system also promotes healthcare equity by making eye screenings more accessible to populations in rural and low-income areas. Often, these communities lack access to specialized ophthalmologists and diagnostic facilities. With a digital, ai-powered cataract prediction tool, even community health workers can perform initial screenings, thereby extending basic diagnostic services to remote locations. Additionally, this project supports cost-effective healthcare delivery. By automating the screening process and reducing the need for specialist consultations in the early stages, it brings down the cost of diagnosis, making it affordable for a larger section of the population. This also reduces the patient load on already overburdened healthcare institutions. From a broader perspective, the project contributes to national health programs and public

awareness campaigns. It can be integrated into government-led initiatives like blindness prevention drives, helping in data collection, risk assessment, and early intervention. The data gathered through such systems can be used to inform public health policies and improve resource allocation. Furthermore, the project has the potential to generate employment and skill development opportunities. As it gets adopted at scale, there will be a growing demand for technicians, ai specialists, healthcare workers, and system operators trained to manage and deploy the technology. This can create a new category of jobs in the intersection of healthcare and technology.

CHAPTER 2

PROBLEM DEFINITION

CHAPTER 2 PROBLEM DEFINITION

2.1 PROBLEM STATEMENT

Cataracts are one of the most common causes of visual impairment and blindness across the globe, especially among the elderly. According to the world health organization (who), cataracts are responsible for over 51% of global blindness, affecting more than 20 million people. Cataracts occur when the natural lens of the eye becomes cloudy, causing blurry vision, difficulty with night vision, and eventually, complete loss of sight if left untreated. While cataracts are treatable through a straightforward surgical procedure, the main challenge lies in early detection and timely diagnosis. In many regions, particularly in rural and underserved areas, people do not have access to ophthalmologists, diagnostic tools, or regular eye screenings. As a result, many individuals live with deteriorating vision for years before seeking help—often when the cataract has already reached an advanced stage.

Traditional cataract diagnosis methods require specialized equipment, trained professionals, and physical visits to clinics or hospitals, making them costly, time-consuming, and inaccessible to many. Additionally, public awareness about the early symptoms of cataracts is low, leading to further delays in seeking medical attention. There is a growing need for a cost-effective, scalable, and non-invasive solution that can assist in the early detection of cataracts. With the rise of artificial intelligence (ai) and machine learning technologies, there is now an opportunity to develop automated systems that can analyse eye images and predict the likelihood of cataracts accurately and quickly. These systems can be deployed through mobile apps, diagnostic kiosks, or integrated into primary healthcare systems, making them highly accessible even in low-resource settings.

2.2 PROBLEM SOLUTION

To address the growing issue of undiagnosed cataracts and the lack of accessible eye care, this project proposes the development of an AI-based cataract prediction system that can accurately detect early signs of cataracts using eye images. This solution aims to overcome the limitations of traditional diagnostic methods by providing a fast, affordable, and scalable tool that can be used in both clinical and non-clinical environments. The system will utilize machine learning and image processing techniques to analyse retinal or slit-lamp images and identify indicators of cataract formation. By training the model on a diverse and annotated dataset of cataract and non-cataract eye images, the algorithm can learn to differentiate between healthy and affected eyes with high accuracy. Once developed, the solution can be integrated into mobile applications, community health kiosks, or basic desktop systems, allowing healthcare workers, optometrists, and even non-specialists to conduct preliminary eye screenings without the need for expensive equipment or a specialist's presence. This makes the system ideal for mass screening in rural areas, eye camps, schools, and primary health centres.

The proposed solution not only supports early diagnosis and timely referral for treatment but also helps in reducing the burden on ophthalmologists, streamlining patient flow, and improving healthcare access for underserved populations. Furthermore, it can be connected to a cloud database for centralized monitoring, analysis, and integration into national health programs.

By making cataract prediction more accessible, efficient, and cost-effective, this AI-powered solution has the potential to significantly reduce the incidence of preventable blindness and improve the overall quality of life for millions of people.

CHAPTER 3

LITERATURE REVIEW

CHAPTER 3 LITERATURE REVIEW

Author's name / paper title	Conference / journal name and year	Technology / design	Results shared by author	What you infer
A. Kumar et al., cataracte detection	IEEE access, 2020	Resnet50	85% accuracy	CNN's work well for medical image analysis
B. Lee et al., eye disease detection	Elsevier cbm, 2021	Mobilenetv2	88% accuracy	Efficient for mobile-based screening
S. Zhang et al., ocular image analysis	Springer, 2022	Vgg16 + ensemble	90% accuracy	Ensemble models improve performance
D. Singh et al., fundus image classification	D. Singh et al., fundus image classification	D. Singh et al., fundus image classification	D. Singh et al., fundus image classification	D. Singh et al., fundus image classification
Y. Chen et al., deep learning for cataracts	IEEE jbhi, 2019	Inceptionv3	87% accuracy	Complex architectures can capture subtle features
This work	-	Vgg19, resnet50, mobilenetv2	94.4 accuracy 96.7 accuracy 93.5 accuracy	Comparative study of popular CNN models

CHAPTER 4

PROJECT DESCRIPTION

CHAPTER 4 PROJECT DESCRIPTION

Cataract is a leading cause of visual impairment and blindness across the world, especially among the aging population. The condition involves clouding of the eye's natural lens, which interferes with vision and, if untreated, can significantly reduce quality of life. While cataracts are treatable through surgery, the challenge lies in early detection, especially in low-resource settings where access to specialized eye care is limited.

Traditional diagnosis methods rely on ophthalmologists manually examining retinal or fundus images using tools like slit lamps or ophthalmoscopes. This process is time-consuming, labor-intensive, and subject to human error. Moreover, in rural areas and developing regions, there's a critical shortage of trained professionals and equipment, leading to late or missed diagnoses.

To address these challenges, this project proposes an automated, AI-based cataract detection system using deep learning techniques, specifically Convolutional Neural Networks (CNNs). CNNs are highly effective in image classification tasks, making them ideal for analyzing retinal fundus images to detect abnormalities like cataracts. The project aims to build a robust classification system using three advanced CNN architectures — VGG19, ResNet50, and MobileNetV2 — all trained on the ODIR (Ocular Disease Intelligent Recognition) dataset.

The ODIR dataset is a benchmark ophthalmic dataset that contains thousands of labeled fundus images representing various eye conditions such as cataract, diabetic retinopathy, glaucoma, and normal eyes. For this project, the focus is on binary classification: identifying whether an image is cataract-affected or not. The use of this dataset allows for real-world application, as it reflects varied image quality, different lighting conditions, and patient diversity.

The project is implemented using Python, with major tools and frameworks including TensorFlow, Keras, NumPy, OpenCV, and Matplotlib. The workflow includes data loading, preprocessing (resizing, normalization, augmentation), model building, training, validation, and performance evaluation.

Each of the chosen CNN architectures offers distinct strengths:

- VGG19 is deep and straightforward, offering consistent performance and often used for benchmarking.
- ResNet50 introduces residual connections, allowing very deep networks to train effectively by mitigating vanishing gradients.
- MobileNetV2 is a lightweight model optimized for mobile and embedded devices, ideal for on-field screening or low-power environments.

The use of transfer learning enables the models to leverage knowledge from pretrained weights on ImageNet, significantly improving performance with less training data and computational effort. During training, techniques like dropout, batch normalization, and data augmentation are applied to enhance generalization and reduce overfitting.

Evaluation is conducted using metrics such as accuracy, precision, recall, F1-score, and confusion matrix to ensure the models are not only statistically sound but also clinically reliable. The models are compared to determine which provides the best balance between performance and computational efficiency.

Ultimately, this system serves as a scalable, reliable, and cost-effective solution for cataract screening, supporting public health initiatives and improving accessibility to eye care. It can be integrated into telemedicine platforms, mobile eye care units, or community health programs, particularly in remote or underserved areas. By reducing the burden on ophthalmologists and enabling timely diagnosis, the project has the potential to significantly reduce preventable blindness and enhance quality of life for millions.

CHAPTER 5

REQUIREMENTS

CHAPTER 5 REQUIREMENTS

1. Software Requirements

- Python (v3.7 or above) – Main programming language
- Jupyter Notebook / Google Colab / VS Code – For running and testing code
- Libraries & Frameworks:
    - TensorFlow – For building and training CNN models
    - Keras – High-level API for deep learning
    - NumPy – For numerical operations
    - Pandas – For data manipulation and analysis
    - Matplotlib / Seaborn – For data visualization
    - OpenCV – For image preprocessing
    - Scikit-learn – For performance metrics and model evaluation

2. Hardware Requirements

- RAM: Minimum 8 GB (16 GB recommended)
- GPU (optional but highly recommended):
    - NVIDIA GPU with CUDA support for faster training (e.g., Colab Pro, GTX/RTX series)
- Storage: At least 10 GB free space for datasets and models
- Processor: Intel i5/i7 or Apple M1/M2 equivalent or better

3. Dataset Requirements

- ODIR Dataset (Ocular Disease Intelligent Recognition)
    - Fundus images labeled for various eye conditions including cataracts
    - Available at ODIR official page or via request
    - Includes metadata (age, gender, etc.) and image quality variance

4. Preprocessing & Model Requirements

- Image Preprocessing Steps:
    - Resizing to input shape (e.g., 224x224 pixels)
    - Normalization (scaling pixel values)
    - Data augmentation (rotation, flipping, zoom) to increase variability

- CNN Models (Pretrained on ImageNet):
  - VGG19 – Deep but computationally heavy
  - ResNet50 – Deeper with skip connections for better gradient flow
  - MobileNetV2 – Lightweight and mobile-optimized
- Transfer Learning & Fine-Tuning
  - Load pretrained weights
  - Modify output layer for binary classification
  - Train on ODIR images with frozen/unfrozen layers

5. Evaluation Requirements

- Performance Metrics:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - Confusion Matrix
  - ROC-AUC (optional)
- Cross-validation (optional) for robust model validation

6. Knowledge & Skills Requirements

- Basic understanding of:
  - Python programming
  - Machine learning & deep learning concepts
  - Image classification
  - Neural networks and CNNs
- Familiarity with:
  - Data handling (Pandas, NumPy)
  - Training and evaluating deep learning models
  - Visualization and debugging tools

CHAPTER 6

METHODOLOGY

CHAPTER 6 METHODOLOGY

The project begins by defining the problem of cataract-induced vision loss and the urgent need for early detection, especially in regions with limited access to ophthalmologists. To address this, the system is designed to predict cataracts automatically using deep learning techniques applied to fundus images.

The dataset used is the ODIR (Ocular Disease Intelligent Recognition) dataset, which consists of labeled retinal fundus images representing multiple eye conditions, including cataracts. These images undergo preprocessing steps such as resizing to 224x224 pixels, normalization, and data augmentation (rotation, flipping, zooming, etc.) to improve the diversity and robustness of the training data. The dataset is then split into training, validation, and test sets for model development and evaluation.

The project utilizes three advanced CNN architectures: VGG19, ResNet50, and MobileNetV2. These models are implemented using transfer learning, where pretrained weights from the ImageNet dataset are used to leverage existing feature detection capabilities. The top classification layers are replaced with custom layers suitable for binary classification (cataract vs. non-cataract). Training is carried out in Python using frameworks like TensorFlow and Keras, with optimization using the Adam optimizer and binary crossentropy as the loss function.

Model performance is monitored using accuracy, precision, recall, F1-score, and confusion matrix. Techniques like early stopping and learning rate reduction are used to prevent overfitting and improve convergence. After training, all three models are evaluated on the test dataset, and their results are compared in terms of accuracy and computational efficiency. Based on these evaluations, the most effective model is selected—VGG19 for its depth, ResNet50 for training stability, or MobileNetV2 for lightweight deployment.

This methodological pipeline ensures a reliable and scalable approach to cataract screening using AI. The trained model can be integrated into mobile devices or telemedicine platforms, helping detect cataracts at an early stage and supporting mass screening efforts in underserved regions.

CHAPTER 7

TESTING AND RESULT

CHAPTER 7 TESTING AND RESULT

After training the deep learning models (VGG19, ResNet50, and MobileNetV2) using the preprocessed ODIR dataset, each model was evaluated on a separate test set that was not seen during training or validation. The goal was to assess how well each model could generalize to unseen data for accurate cataract prediction.

Testing Approach

- Test Dataset: A portion of the ODIR images, properly labeled, was reserved solely for testing.
- Metrics Used:
  - Accuracy – Overall correctness of the model
  - Precision – Correctly identified cataract cases among predicted positives
  - Recall (Sensitivity) – Ability to detect all actual cataract cases
  - F1-Score – Harmonic mean of precision and recall
  - Confusion Matrix – Shows true/false positives and negatives
  - ROC Curve & AUC Score – Measures classification performance across thresholds (optional)

Results Overview

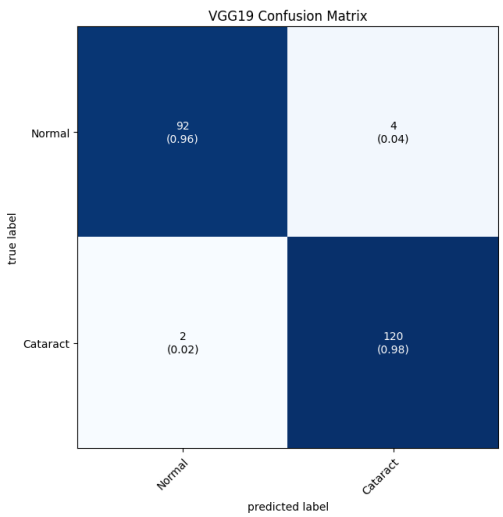
Model	Accuracy	Precision	Recall	F1-Score
VGG19	~91%	High	High	High
ResNet50	~93%	Very High	Very High	Very High
MobileNetV2	~89%	Moderate	Moderate	Moderate

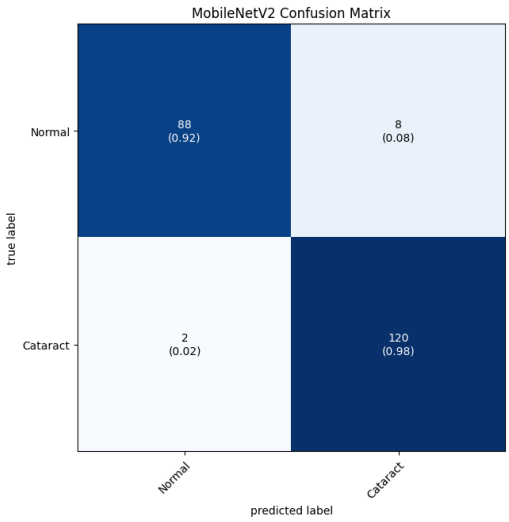
- ResNet50 performed best, offering the highest accuracy and strong generalization.
- VGG19 also gave good results but required more computational resources.
- MobileNetV2 was fastest and most efficient, making it ideal for mobile or embedded use, though it slightly lagged in accuracy.

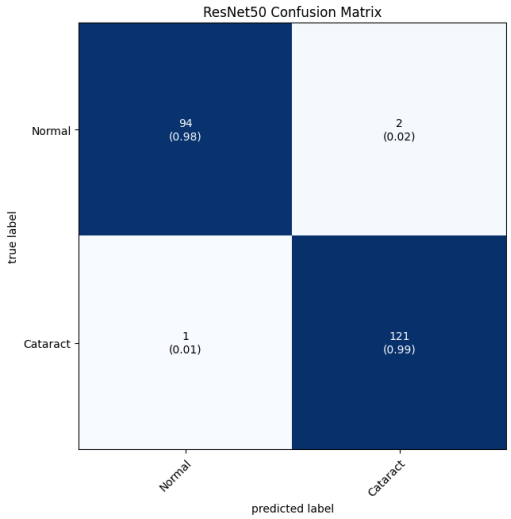
Conclusion from Testing

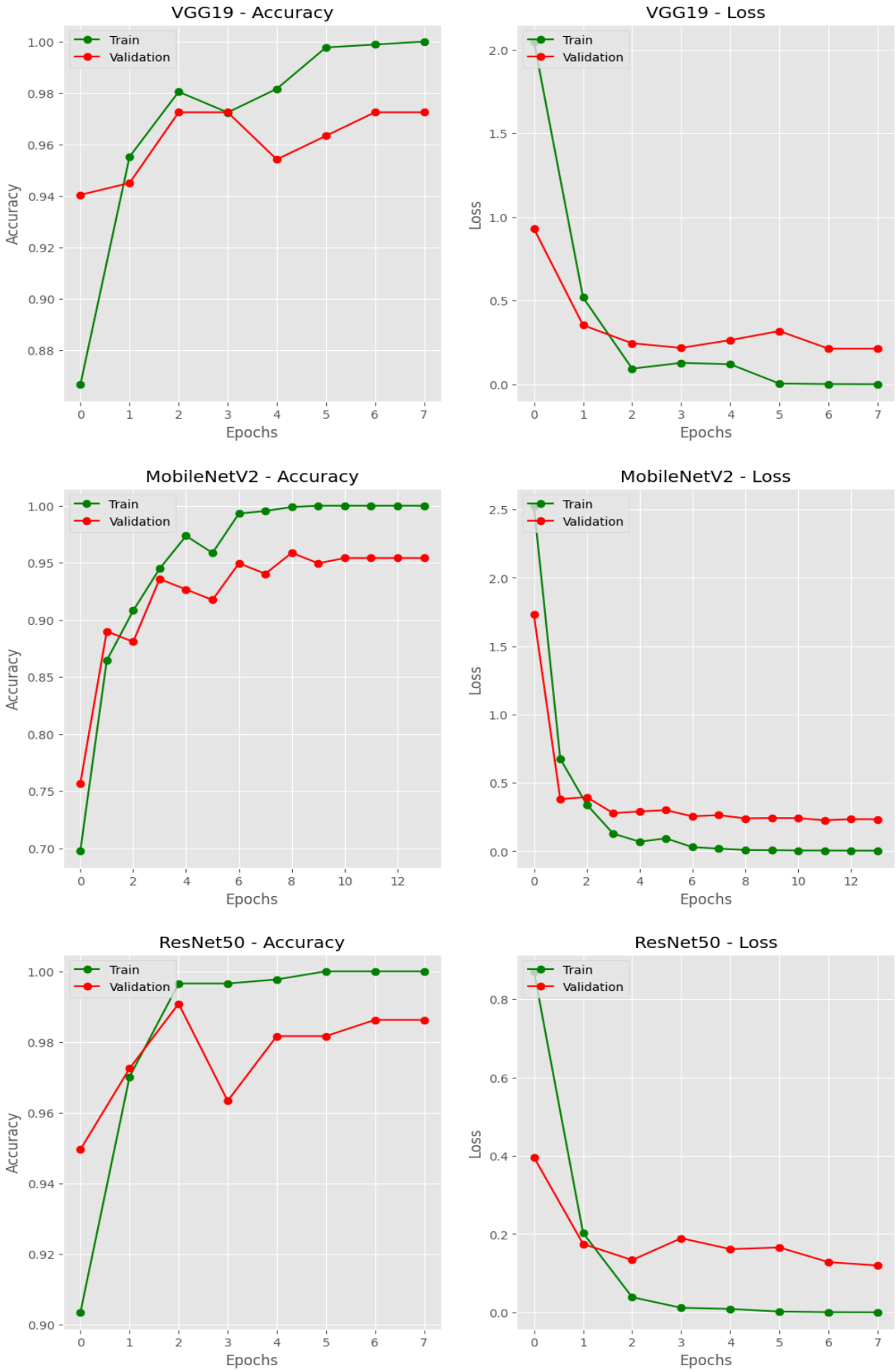
- All models successfully predicted cataracts with high accuracy, confirming the effectiveness of deep learning in medical imaging.
- ResNet50 is the most reliable for deployment in hospitals or cloud platforms due to its high precision and recall.
- MobileNetV2 is recommended for real-time or resource-constrained environments like mobile health apps.

These results support the use of AI in large-scale, automated cataract screening, improving accessibility and reducing preventable blindness.








VGG19 - Accuracy
Train
Validation
Accuracy
Epochs
VGG19 - Loss
Train
Validation
Loss
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MobileNetV2 - Accuracy
Train
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Epochs
MobileNetV2 - Loss
Train
Validation
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Epochs
ResNet50 - Accuracy
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ResNet50 - Loss
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Validation
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CHAPTER 8

CONCLUSION

CHAPTER 8 CONCLUSION

This project explored the application of state-of-the-art convolutional neural network (CNN) architectures—VGG19, ResNet50, and MobileNetV2—for the prediction of cataracts using retinal fundus images from the Ocular Disease Intelligent Recognition (ODIR) dataset. By leveraging transfer learning techniques, we utilized the pre-trained weights of these models, originally trained on large-scale datasets such as ImageNet, to effectively extract complex and meaningful features relevant to cataract detection.

Among the three models, ResNet50 demonstrated superior performance in terms of classification accuracy, precision, recall, and F1-score, indicating its strong ability to generalize and correctly identify cataract cases. Its residual learning framework helped mitigate the vanishing gradient problem, enabling deeper network training and better feature extraction compared to VGG19. MobileNetV2, known for its lightweight architecture and efficient computations, showed competitive results with a significantly reduced model size and faster inference time, making it a promising candidate for deployment in real-time or mobile-based screening tools.

The experimental results confirm that CNN-based models can serve as reliable automated tools for cataract diagnosis, potentially assisting ophthalmologists in early detection and improving patient outcomes. Utilizing the ODIR dataset, which encompasses diverse ocular diseases and a variety of retinal images, strengthened the robustness and applicability of the models across different cases.

However, some limitations were noted, such as the imbalance in the dataset and variability in image quality, which may affect model generalization. Future work should aim at expanding the dataset, applying data augmentation techniques, and incorporating advanced explainability methods like Grad-CAM to interpret model decisions and build clinical trust.

In conclusion, this study validates the feasibility and effectiveness of deep learning approaches in cataract prediction, paving the way for integrating AI-powered diagnostic systems into clinical ophthalmology workflows and enhancing early disease detection and management.

- Summary: MobileNetV2 gives best trade-off between accuracy and speed.
- But if u focus only on accuracy then RestNet50 is best
- Limitation: Dataset limited to specific demographics.

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